

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit II | Session 4 | SLO 1: Abstract Classes & Interfaces (Code: 2041)

Q: Activity: Design a banking system illustrating the difference between an Abstract Class and an Interface. Implement an abstract class BankAccount and an interface Transaction.

Ans: Java Implementation:

```
interface Transaction {
    void deposit(double amount);
    void withdraw(double amount);
}

abstract class BankAccount implements Transaction {
    String accountNumber;
    double balance;

    BankAccount(String accNo, double bal) {
        this.accountNumber = accNo;
        this.balance = bal;
    }

    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: $" + amount + ", New Balance: $" + balance);
    }

    abstract void applyInterest();
}

class SavingsAccount extends BankAccount {
    SavingsAccount(String accNo, double bal) {
        super(accNo, bal);
    }

    public void withdraw(double amount) {
        if (balance >= amount) balance -= amount;
    }

    void applyInterest() {
        balance += balance * 0.04;
        System.out.println("Interest applied. Balance: $" + balance);
    }
}

public class BankDemo {
    public static void main(String[] args) {
        SavingsAccount sa = new SavingsAccount("SA-101", 1000.0);
        sa.deposit(500.0);
        sa.applyInterest();
    }
}
```